TECHNICAL REPORT

Axial Compressor Rotor Modeling & Meshing

Structured hexahedral mesh generation and parametric mesh-quality assessment

Study item	Documented configuration
Software	ANSYS Workbench / BladeGen / TurboGrid 2026 R1 Student
Automation	Python 3.11; ansys-turbogrid-core 0.9.4
Domain	One rotor blade passage; 24 blade sets in source geometry
Experimental matrix	Baseline plus four single-parameter variations
Data source	5 TurboGrid Mesh Statistics CSV exports

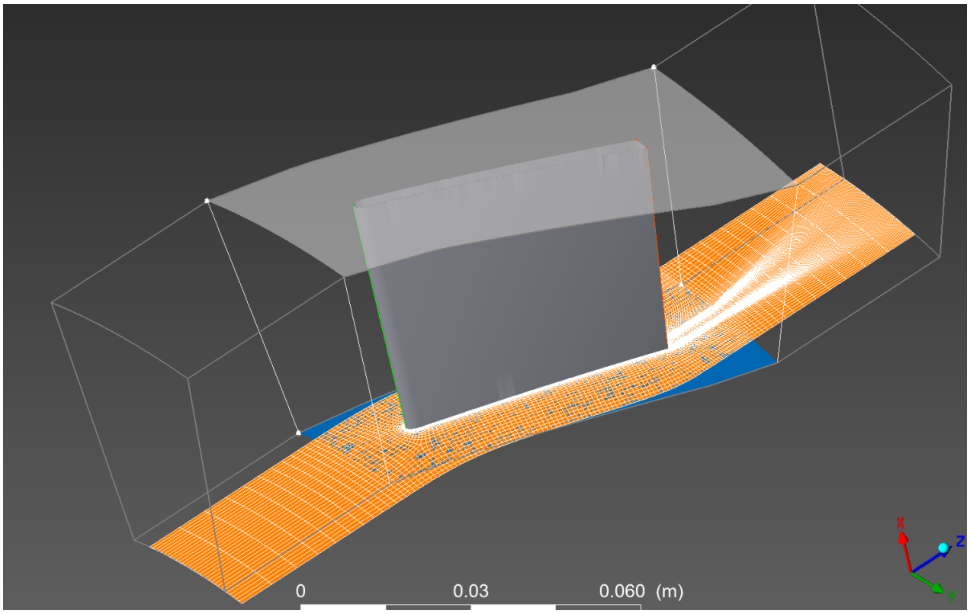


Figure 1. Single blade passage and representative structured mesh view (ANSYS TurboGrid capture).

Abstract. This report documents the creation of an axial-compressor rotor blade-passage model in ANSYS BladeGen and the generation of a structured hexahedral mesh in ANSYS TurboGrid. Four parameter variations were assessed against a baseline using the exported mesh-quality statistics. All reported cases satisfy the project-specific TurboGrid “Bad” limits. Run 2 offers the lowest intermediate orthogonality percentage (0.297913%) and maximum edge-length ratio (137.487); Run 3 offers the lowest intermediate connectivity percentage (0.0434401%). These measurements characterize geometric mesh quality only; no CFD solution or mesh-independence claim is made.

Keywords: axial compressor; rotor; BladeGen; TurboGrid; structured mesh; orthogonality; parametric study.

1. Introduction and study scope

The objective is to create a reproducible, traceable workflow for modeling a simplified axial-compressor rotor and assessing structured mesh geometry in a single blade-to-blade passage. The study addresses geometric definition, multi-block mesh generation, quality checks, and the sensitivity of selected quality indicators to meshing parameters. It does not model the flow field or aerodynamic performance.

1.1 Rotor geometry and computational domain

Quantity	Recorded value
Rotor blade-set count	24
Hub radius	100 mm
Shroud radius	150 mm
Blade span	50 mm
Reported axial blade extent	45 mm
Axis of rotation	Z
Geometry units	millimetres (mm)

Geometric values are taken from the project README and BladeGen export metadata (*BG.inf*). The domain consists of a circumferential passage bounded by neighboring periodic surfaces, a hub wall, a shroud boundary, and the blade surfaces.

1.2 Modeling-to-meshing workflow

Stage	Action	Evidence
BladeGen	Generate rotor meridional and blade geometry	geometry/BG.bgd
Export	Prepare hub, shroud, profile curves and machine data	states/BG_*.curve and BG.inf
TurboGrid	Create ATM-based structured hexahedral passage mesh	states/*.tst
Analysis	Compute mesh statistics; export CSV through Python	scripts/*.py and results/*.csv
Comparison	Compare Baseline and four independent perturbations	results/*.xlsx

The machine-data file identifies a right-handed coordinate system and one blade per set. BladeGen files and derived curve files are retained in the repository for traceability. The TurboGrid state files are not independent geometry archives: they may require relinking to these inputs in another environment.

2. Baseline mesh generation

The baseline mesh was generated using TurboGrid's automatic turbomachinery topology and a global size factor of 1.0. Mesh density was controlled by spanwise passage resolution, boundary-layer refinement and the target maximum expansion rate. The saved state was subsequently loaded and the topology unsuspended through PyTurboGrid before the statistics were exported.

Baseline mesh statistic	Reported value
Total nodes	223,857
Total elements	205,244
Spanwise blade elements (state setting)	26
Global size factor	1.00
Boundary-layer proportional factor base	3.00
Target maximum expansion rate	1.30

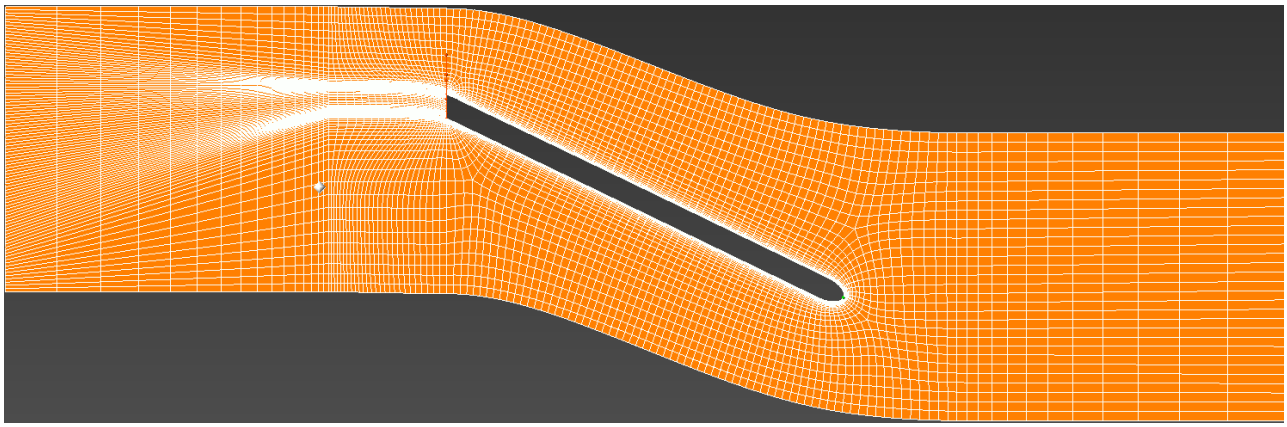


Figure 2. Blade-to-blade mesh view showing the blade profile and the surrounding structured cells (TurboGrid capture).

The representative cross-section shows mesh concentration around the blade contour and transitions toward inlet and outlet portions of the passage. A visual plot supports inspection of local topology but cannot replace a numerical assessment of element quality.

3. Quality assessment criteria

TurboGrid reports a value and three classification percentages for each mesh measure: % Bad, % ok (intermediate), and % OK. These percentages are evaluated separately for each measure, under the current Mesh Limits configuration. Consequently, "0% Bad" indicates no violations of the configured limits, not zero discretization error.

3.1 Baseline mesh statistics

Measure	Reported value	% ok	% Bad
Orthogonality angle	37.9389 deg	0.327441	0.000000
Minimum face angle	35.1806 deg	0	0
Maximum face angle	144.7372 deg	0	0
Edge length ratio	200.302	0	0
Element volume ratio	4.50705	0	0
Minimum volume	2.89777×10^{-13} m^3	0	0
Connectivity number	10	0.044671	0
Skewness	0.609138	0	0

Table 1. Baseline quality measures. Angular outputs originally expressed in radians were converted to degrees (1 rad = 180/pi deg); the minimum cell volume was converted from cm^3 to m^3 (1 cm^3 = 10^{-6} m^3).

For the Baseline, the largest edge-length ratio is 200.302; the reported skewness is 0.609138 and the minimum cell volume is positive. Only Orthogonality Angle and Connectivity Number have non-zero intermediate-category percentages. The configured upper limits for Edge Length Ratio and Skewness are 1,000 and 0.90, respectively; this report preserves the configured limits rather than adjusting them to optimize pass rates.

3.2 Interpreting the percent classification

The % ok column must not be confused with the quality-angle value: reducing *Orthogonality % ok* is beneficial only when those cells transfer into the % OK category, without producing new Bad elements or deterioration elsewhere. A small change in the percentage for Connectivity Number may also arise from a change in the total node count, rather than a change in the largest connectivity value.

The exported CSV values are rounded by TurboGrid. Reported percentage comparisons below should therefore be interpreted as small changes in classification fractions, not as a direct measure of CFD accuracy.

4. Experimental design: parameter perturbations

Four independent one-factor changes were compared with the same baseline. This design allows a first-order qualitative attribution of each observed change to the modified control within the tested range; it does not quantify parameter interactions or establish a global optimum.

Case	Passage factor	BL factor base	Expansion rate	Spanwise blade elems.
Baseline	1.00	3.00	1.30	26
Run 1	1.15	3.00	1.30	32
Run 2	1.30	3.00	1.30	39
Run 3	1.00	3.30	1.30	26
Run 4	1.00	3.00	1.25	26

Table 2. One-factor-at-a-time configurations. Parameters and spanwise blade element counts were verified against the saved TurboGrid state files.

4.1 Orthogonality classification response

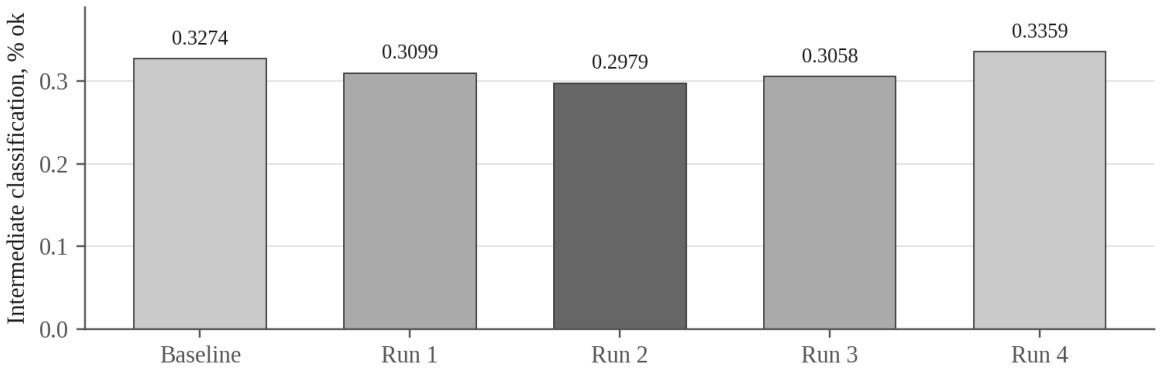


Figure 3. Intermediate-category orthogonality percentage for each configuration (lower is preferable when % Bad remains 0).

Run 2 has the smallest intermediate Orthogonality share (0.297913%), a reduction of 0.029528 percentage points, or approximately 9.02% relative to the Baseline. In contrast, Run 4 increases this share to 0.335912%. The associated maximum orthogonality-angle values vary slightly across the cases, and should be read together with the classification percentages.

5. Mesh-quality comparison and discussion

Case	Ortho % ok	Connect % ok	Max edge ratio	Volume ratio	Skewness
Baseline	0.327441	0.044671	200.302	4.50705	0.609138
Run 1	0.309938	0.045321	171.599	4.01011	0.609138
Run 2	0.297913	0.045833	137.487	3.70987	0.609138
Run 3	0.305818	0.043440	232.031	4.63521	0.609140
Run 4	0.335912	0.045640	201.512	4.46753	0.609137

Table 3. Comparison extracted from the five CSV exports. All eight quality measures report 0% Bad for all five configurations.

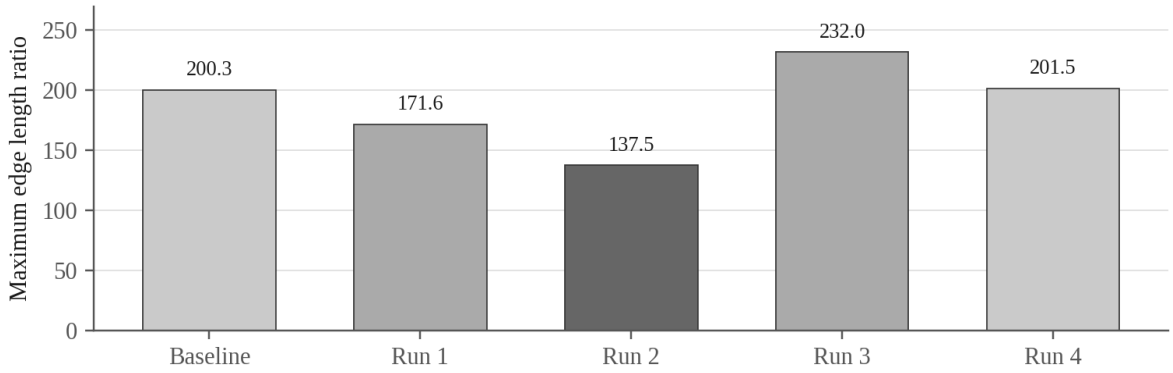


Figure 4. Maximum edge-length ratio for the five meshing configurations.

Run 1 and Run 2. Increasing the passage spanwise count factor from 1.00 to 1.15 and 1.30 corresponds to more spanwise blade elements (26, 32, 39) and progressively lower maximum edge-length ratios (200.302, 171.599, 137.487). The Run 2 reduction is 31.36% versus Baseline, accompanied by the smallest orthogonality intermediate share.

Run 3. Increasing the boundary-layer factor base to 3.30 yields the smallest intermediate Connectivity share, 0.0434401% (approximately 2.76% lower relative to Baseline), but the maximum edge-length ratio rises to 232.031. The maximum connectivity number remains 10 across the runs, limiting the strength of any topology-improvement conclusion.

Run 4. Reducing the target maximum expansion rate to 1.25 does not improve the selected maximum edge-length ratio over Baseline (201.512 versus 200.302), and the intermediate Orthogonality percentage rises. It may influence local boundary-layer transitions, but a local surface map would be required to substantiate that mechanism.

6. Local blade-mesh structure

A blade-to-blade view offers a qualitative check of cell alignment near the blade edges. The following crops come from the same TurboGrid cross-section supplied during project development. They are visual documentation, not independent quantitative quality metrics.

6.1 Leading-edge vicinity

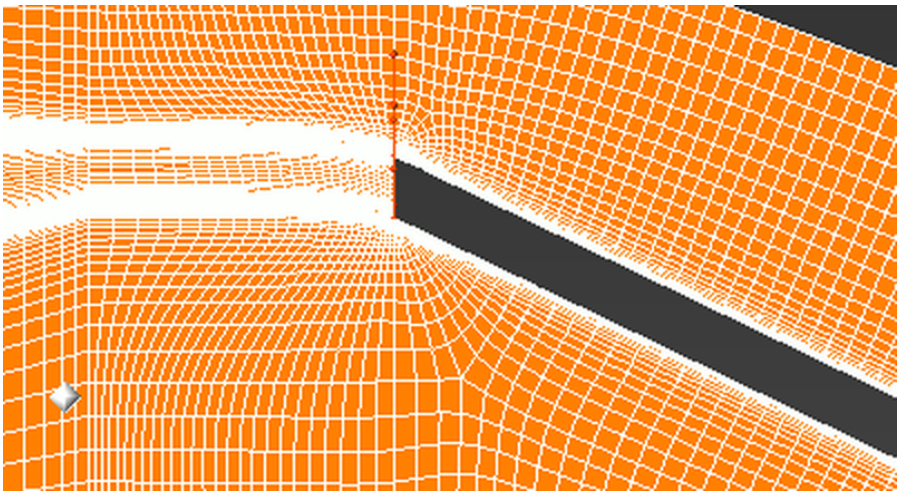


Figure 5. Representative mesh concentration near the blade leading edge; selected region of the blade-to-blade plot.

6.2 Trailing-edge vicinity

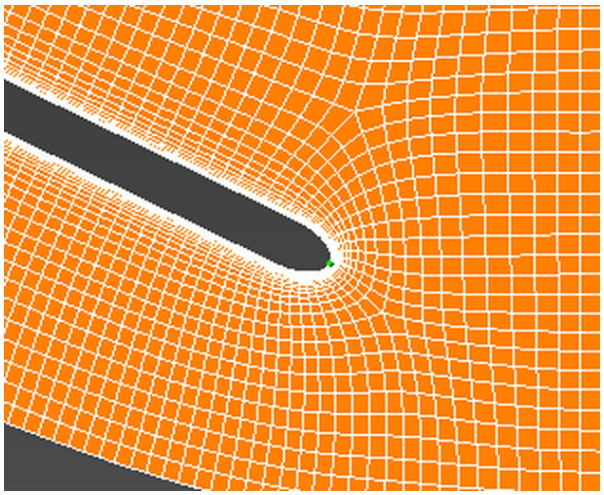


Figure 6. Structured cells near the trailing edge and downstream wake region in the representative cross-section.

The apparent continuity of cell lines is promising for a preliminary geometry-focused assessment. However, images alone cannot establish numerical robustness; local orthogonality, edge-length ratio, skewness and volume statistics must remain the acceptance criteria.

7. Reproducibility and limitations

The repository contains BladeGen geometry, machine-data and curve files, five TurboGrid state files, a Python exporter and the resulting mesh-quality CSV files. The saved state can refer to absolute file paths; consequently, a clean-checkout test should confirm geometry relinking and mesh regeneration before claiming full portability.

7.1 Reproduction protocol

Step	Command or action
1	Install ANSYS TurboGrid 2026 R1 with an appropriate license.
2	Install Python 3.11 and run: python -m pip install -r requirements.txt
3	Open a terminal in the repository root directory.
4	Run: python scripts/export_stats.py
5	If needed, relink the .curve / BG.inf files from states/ in TurboGrid.
6	Compare the five generated CSV outputs to the committed values.

7.2 Limitations

The study does not include flow boundary conditions, pressure ratio, efficiency, mass flow, turbulence-model details, wall-resolution targets, CFD residuals or grid-convergence results. The total nodes/elements of Run 1-4 were not present in the five exported quality-statistics CSV files and are therefore not reported here. Selection of the least expensive acceptable mesh is not possible from these data alone.

8. Conclusions

A structured 3D rotor-passage mesh was created successfully and evaluated for five saved configurations. Within the configured TurboGrid criteria, all cases have 0% Bad values across the eight mesh-quality measures. Run 2 is a useful candidate for subsequent work because it combines the lowest intermediate Orthogonality share with the lowest maximum edge-length ratio in the tested set. Run 3 reduces the intermediate Connectivity share marginally but worsens edge-length ratio. No single scalar quality measure can substitute for a full mesh assessment, and no aerodynamic performance claims are made.

Next steps are to record node/element totals for each run, verify the clean-checkout regeneration of all cases, assess local quality contours for the selected configuration, and only then perform a CFD-based grid-convergence study if the project scope expands.

Data provenance and references

[1] Project repository and source files: https://github.com/HungDNVu/axial-compressor-rotor-modeling-meshing.
[2] Statistics: *results/Baseline_stats.csv* and the four *Run*_stats_stats.csv* files; state settings: *states/*.tst*; machine metadata: *states/BG.inf*.
[3] Project visualization: user-supplied TurboGrid screenshots from the modeling and mesh-quality review; analysis and charts in this document computed from the original CSV values.